

Design and Implementation of 64 Bit Multiplier using Vedic Algorithm

Amish Jais, Prasanna Palsodkar, *SM-IEEE*

Abstract—As floating point architecture is very hot topic for researchers so challenges are always there to design the efficient Floating point architecture. Out of other operations, Floating point multiplication is the most commonly used operation and it requires the multiplication of the mantissa of Floating point numbers. This paper presents the highly efficient 64 bit multiplier for the mantissa calculation using rule or sutra of Vedic mathematics called Urdhva Tiryakbhyam Sutra which deals with vertically and crosswise multiplication. Using this sutra in the computation algorithm of DSP processors, can enhance the efficiency and at the same time can reduce the complexity, area, power consumption and delay. Starting from the design of 2 bit Vedic multiplier we went up to design 64 bit Vedic multiplier as presented in this paper. Vedic multiplier is coded in Verilog HDL and targeted to three different families of FPGA Spartan6, Virtex5 and Virtex6 in Xilinx 13.1 ISE software. Result is compared with the Karatsuba, Vedic-Karatsuba and Optimized Vedic multiplier and found 33% reduction in delay.

Index Terms—FPGA, Urdhva Tiryakbhyam Sutra, Vedic Algorithm, Vedic Multiplier

I. INTRODUCTION

Floating point arithmetic is a very important topic for researchers as floating point numbers are widely used in many application and as compared to a fixed point number it has better resolution and accuracy. The 64 bit IEEE 754 floating point number format, shown in figure 1, is divided into three blocks i.e. sign field, exponent field and mantissa field. Sign field is of '1' bit, exponent field is of '11' bit and the mantissa field is of '52' bit. The operation of sign field and the exponent field is quiet easy. The mantissa field is one of large field and involves the multiplication operation of two 52 bit operands [1].

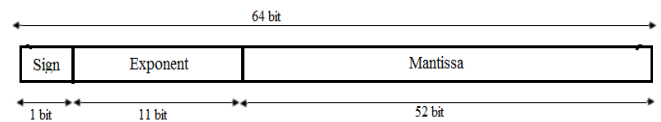


Fig. 1. 64 Bit IEEE 754 floating point number format [1]

The operation of sign field and the exponent field is quiet easy. The mantissa field is one of large field and involves the multiplication operation of two 52 bit operands [1].

There are four major Arithmetic operations: addition, subtraction, division and multiplication. As compared to other three operations, multiplication is the slowest operation. Multiplier is the most important blocks of various DSP processors, microprocessors, and various FIR filters. As multiplication requires large amount of execution time of most of the DSP application, therefore there is always a need to have high speed and low delay multipliers. An optimized time delay is the important requirement for number of application as well. Earlier multiplication had a addition and subtraction followed by shifting the operands. Vedic multiplication technique is far different than these algorithms. Earlier designed 64 bit Vedic Multiplier had a larger delay and lower speed and are not very convenient to work with, therefore we had decided to reduce the delay and incresing the speed of the multiplier so that it can be used in any application very conviniently.

Section I consist of introduction which gives the brief idea about floating point followed by the Related work comes under Section II. Section III includes the Methodology which tells the basic process of Urdhva Tiryakbhyam Sutra. Section IV gives the Design and Implementation of different modules starting from 2 bit multiplier to 64 bit multiplier. Section V and VI contains the Experimental Results and Conclusion respectively.

II. RELATED WORK

V. budhe, P.M.Palsodkar, P.P.Palsodkar, in their paper, "Design and Verification of Dadda Algorithm based Binary Floating Point Multiplier", had developed a fast 32 bit floating point multiplier. In this paper they had used Dadda algorithm for the multiplication of the mantissas of two inputs. They minimize the delays in arithmetic operations at every stage in order to improve the speed of multiplier [1].

P. Singh, B. Kakani, in their paper, "Performance Comparison of Floating point multipliers by using different multiplication algorithm", had used Vedic algorithm which is fast method based on ancient Indian mathematics for mantissa multiplication. Since mantissa multiplication is the important

Amish Jais, M. Tech (student), Department of Electronics Engineering, Yeshwantrao Chavan College of Engineering, an autonomous institution, affiliated to RTMNU, Nagpur, India(e-mail: amish041190@gmail.com)

Prasanna Palsodkar is with Department of Electronics Engineering, Yeshwantrao Chavan College of Engineering, an autonomous institution, affiliated to RTMNU, Nagpur, India (e-mail: palsodkar.prasanna@gmail.com)

aspect of the floating point multiplication, therefore we kept our focus to design a multiplier. Compared to the Dadda algorithm Vedic algorithm proves to be a faster [2].

Jagadguru Swami Sri Bharati Krisna Tirthji Maharaja, in their paper, “Vedic Mathematics” had given 16 sutras and 13 Upya sutras after thorough research in Atharva Veda. Urdhva Tiryakbhyam means “Vertically and Crosswise” multiplication and subsequent partial products addition [3].

R. K. Kodali, L. Bopanna, S. S. Yenamachintala, in their paper, “FPGA implementation of Vedic floating point multiplier”, had developed a Vedic multiplier to perform both 32 bit and 64 bit format floating point operations and its performance had been compared with Booth algorithm and Karatsuba based floating point multipliers. They had also shown that based on the device utilization and performance comparison, Vedic multiplier outperforms Karatsuba multiplier both for 32 bit and 64 bit format [4].

Ramalatha, M.Dayalan, P Priya, K D Dharani and S Deoborah, in their paper, “High speed energy efficient ALU Design using Vedic multiplication techniques”, had designed high speed ALU using Vedic algorithm technique. The comparative analysis of Karatsuba algorithm, Vedic-Karatsuba algorithm and Optimized Vedic multiplier for 8, 16, 32, and 64 bit was carried out in their paper. They had found the reduced delay for Vedic multiplier thus proving the efficiency of Vedic multiplier for all bit size over other algorithms i.e. Karatsuba and vedic-Karatsuba [5].

Garima Rawat, , Siddharth Goyal, Shefali Kala, Khyati Rathore and Poornima Mittal, in their paper, “Design and Analysis of ALU: Vedic Mathematics Approach”, had designed the Arithmetic and Logical Unit. By keeping the importance of a multiplier in mind they had designed 8 bit multiplier and also presented the comparison between Array multiplier, Booth multiplier and Vedic multiplier. They got the reduction in delay for Vedic multiplier against the Array and Booth multiplier for 8 bit size [6].

Pushpalata Verma, K. K., Mehta, in their paper, “Implementation of an Efficient Multiplier based on Vedic Mathematics Using EDA Tool”, had developed the 8 bit multiplier by using vertically and crosswise approach which is one of the 16 sutras of the Vedic algorithm. They had compared the obtained results with Booth multiplier and the Array multiplier on the grounds of path delay. They found that the multiplier based on Vedic algorithm is faster than Booth and Array multiplier [7].

Honey Durga Tiwari, Chan Mo Kim, Ganzorig Gankhuyag and Yong Beom Cho, in their paper, “Multiplier design based on ancient Indian Vedic Mathematics”, had investigated Vedic multiplication formulae, Urdhva tiryakbhyam and Nikhilam in detail. They found the large propagation delay in case of large number while dealing with the Urdhva tiryakbhyam sutra,

therefore they suggested the Nikhilam sutra for the large numbers [8].

S.S.Mahakalkar, S.L.Haridas, in their paper, "Design of High Performance IEEE754 Floating point multiplier using Vedic mathematics", had done a synthesis and verification of 32 bit and 64 bit IEEE 754 high speed Floating point multiplier using Urdhva Tiryakbhyam sutra. Carry propagation delay is reduced by using carry save adder while implementation of four (27 x 27 bit multiplier for 64 bit) and (12 x 12 bit multiplier for 32 bit) in order to achieve high speed [9].

Prabir Saha, Partha Bhattacharyya, Arindam Banerjee, Anup Dandapat, in their paper, “High Speed ASIC Design of Complex Multiplier Using Vedic Mathematics”, had developed the 8 bit and 16 bit multiplier by using Vedic technique and also calculated the performance parameters like dynamic power consumption and propagation delay. They had compared the proposed design with the previously reported complex multipliers and found 25% improvement in speed [10].

III. METHODOLOGY

A. Vedic Multiplication Technique

The spiritual leader and the mathematician Sri Bharti Krisna Tirthaji had reconstructed 16 sutras or fundamental principles and 13 sub-sutras or corollaries. Vedic mathematics is far simpler than the conventional mathematics and it understand from the fact that all the calculations are carried out mentally and it is not necessary to write down each and every steps. The word Vedic is derived from the word ‘Veda’ which literally means the store house of all knowledge [3].

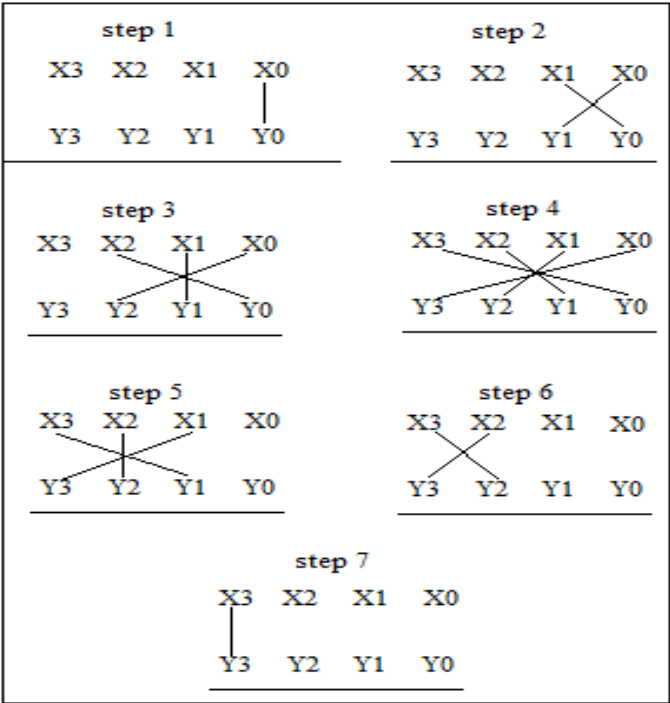


Fig. 2. 4 Bit Vedic Multiplication line diagram.

B. Urdhva Tiryakbhyam Sutra

The 'Urdhva Tiryakbhyam Sutra' is one of that 16 sutra which used for the multiplication and that sutra is used in this paper. 'Urdhva' means 'vertically' and 'tiryakbhyam' means 'crosswise'. Fig. 2 shows the Line Diagram for 4 Bit Multiplication. It is an universal formula which is applicable to all kind of multiplications. The algorithm is compatible for multiplication of any two numbers representing bit length of n. With this, it introduces a parallel execution of a partial product and sums. The clock frequency is not a hurdle in accounting the calculations. This technique shows the sign of advantages while dealing with large delays and complexity of conventional multipliers [3].

IV. DESIGN AND IMPLEMENTATION

A. 2*2 Bit Multiplier

The scheme shown below is for 2 bit numbers A and B where $A=a_1a_0$ and $B=b_1b_0$. Process starts with the vertical multiplication of LSB's resulting to the LSB of the final product. Next is the multiplication of LSB of multiplicand and the next higher bit of multiplier followed by the addition with the product obtained after the multiplication of LSB of multiplier and the next higher bit of multiplicand i.e. crosswise. This summation gives the second bit of end product and the carry is added to the previous product acquire by multiplying the MSB's, which leads the sum and carry terms. The sum and carry act as corresponding third and fourth product term respectively.

$$S_0 = a_0b_0 \quad (1)$$

$$C_1S_1 = a_1b_0 + a_0b_1 \quad (2)$$

$$C_2S_2 = C_1 + a_1b_1 \quad (3)$$

From the above equations, the final product is $C_2S_2S_1S_0$. The implementation of 2*2 bit vedic multiplier is been done using four AND gates alongwith two half adders.

B. 4*4Bit Multiplier

The multiplication of 4*4 bit number using Urdhva Tiryakbhyam Sutra is illustrated below. The processing of 4*4 bit multiplier is depicted in Fig. 3. Let us consider the multiplication of two binary numbers $a_3a_2a_1a_0$ and $b_3b_2b_1b_0$. The output is obtained in 8 bit length say S_7-S_0 . Now the numbers a and b is bifercated into two equal parts of two bits $aH-aL$ and $bH-bL$. The 8 bit end product term is given as:

$$P = a * b = (aH-aL) * (bH-bL) = aH * bH + (aL * bH + aH * bL) + aL * bL.$$

This 4*4 bit multiplication is implemented using four 2 bit multipliers and a ripple carry adder of 4 bit.

C. N*N Bit Multiplier

The multiplier of N*N bit number using vedic algorithm is as shown in Fig. 4. As we have discussed earlier about 4*4 bit multiplier, both the operands are required to divide in two

parts i.e. higher and lower part and then the multiplication is carried out vertically and crosswise manner.

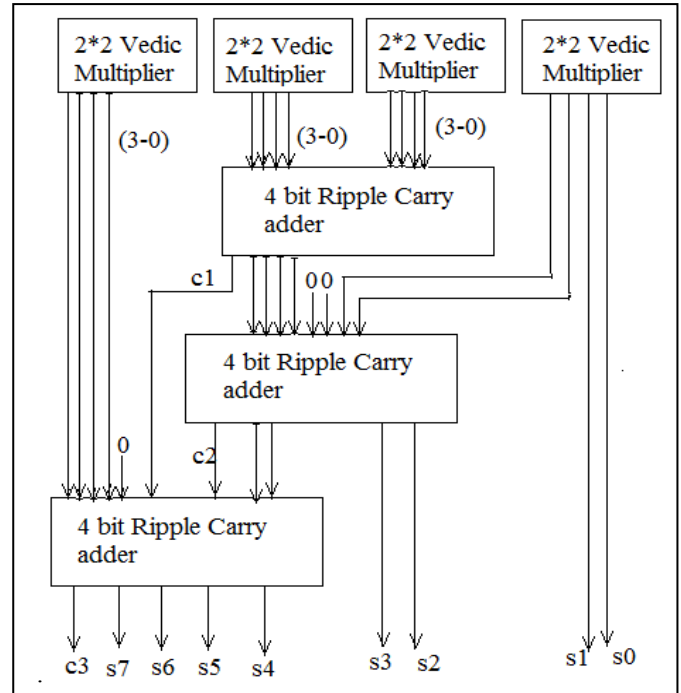


Fig. 3. Schematic of 4*4 bit vedic multiplier.[5]

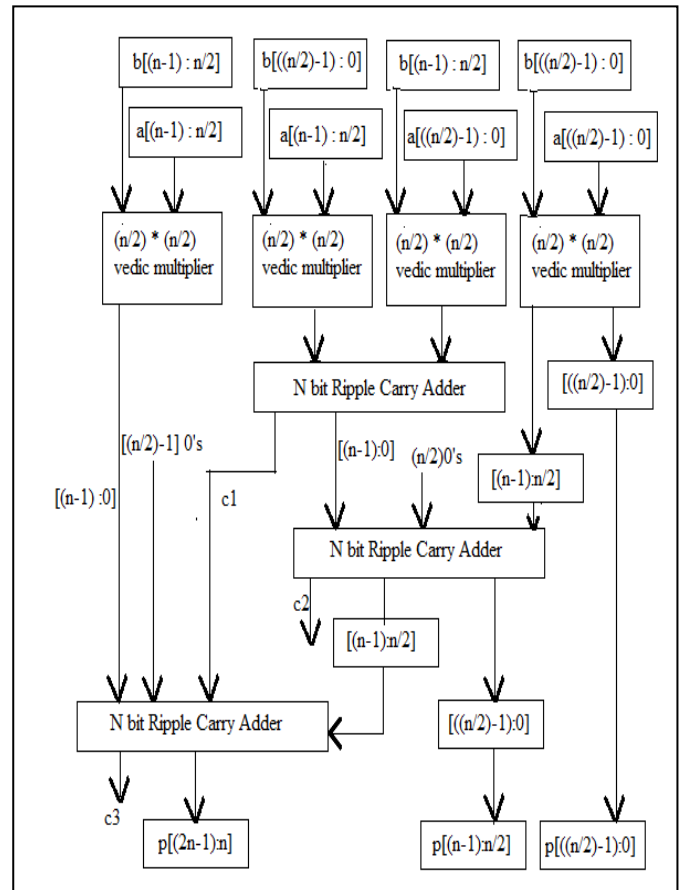


Fig. 4. Schematic of N*N bit Vedic multiplier [6].

D. 64*64 Bit Multiplier

From the available multiplier of 4*4 bit number we have developed multiplier of 8*8 bit, then multiplier of 16*16 bit, multiplier of 32*32 bit and finally we have developed 64*64 bit multiplier. The simulation results of all the multipliers are presented in the later sections. The drawn N*N bit Vedic multiplier gives us the direct idea to make 64*64 bit multiplier simply by putting 'N=n=64' in the architecture shows in Fig. 5.

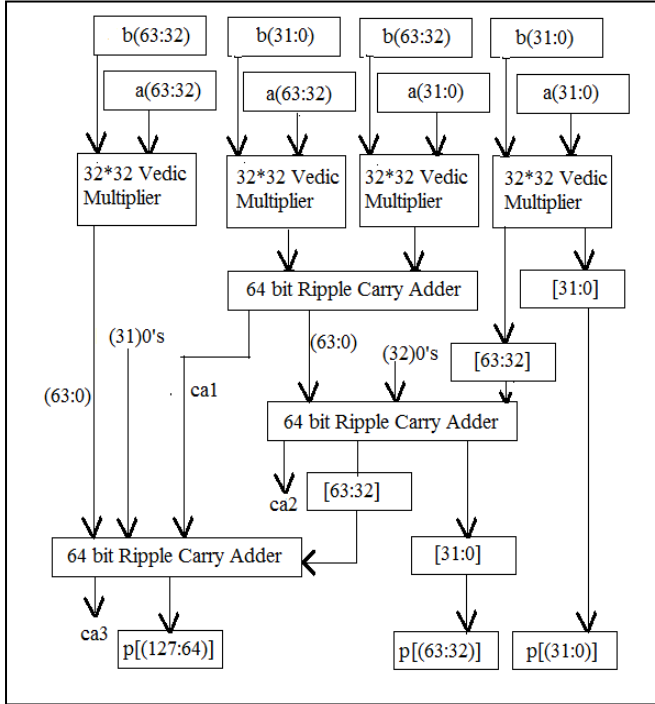


Fig. 5. Schematic of 64*64 bit Vedic multiplier.[6]

V. EXPERIMENTAL RESULTS

The simulation of proposed 64 bit multiplier has been done to calculate the lowest path delay for different multipliers. This section also includes the comparative result analysis of Vedic multiplier and previously designed multipliers. The table I shows the area utilization of proposed Vedic multiplier. Table II shows the delay analysis of proposed 64 bit Vedic multiplier targeted for different FPGA's.

TABLE I.
DEVICE UTILIZATION REPORT OF PROPOSED 64 BIT
MULTIPLIER

LOGIC UTILIZATION			
	USED	AVAILABLE	UTILIZATION
Number of slice LUT's	2622	150720	1%
Number of fully used LUTs-ff pairs	0	2622	0%
Number of bonded IOB's	128	600	21%

TABLE II.
DELAY ANALYSIS OF PROPOSED MULTIPLIER FOR DIFFERENT
FPGA.

DELAYS IN NS			
Module	Virtex-5 ML510	Spartan-6 SP605	Virtex-6 ML605
64 bit Multiplier (Proposed)	21.985 ns	24.932 ns	15.770 ns

Additionally, to prove the efficiency of our proposed Vedic mathematics based 64 bit multiplier, the synthesized report and simulation results has been compared with the previously designed 64 bit multiplier using a same algorithm. Table III shows the efficiency of proposed multiplier of 64 bit. It shows that the proposed multiplier has the lowest combinational path delays in comparison with others.

TABLE III.
DELAY COMPARISON FOR DIFFERENT ALGORITHMS

Bit Size = 64 Bit				
Algorithm	Karatsuba [5]	Vedic- Karatsuba [5]	Optimized Vedic [5]	Propo- sed Vedic
Delay (ns)	150.922	92.448	44.87	15.770

VI. CONCLUSION

A very effective method, i.e. Urdhva-Tiryakbhyam Sutra based on Vedic mathematics is used in this paper for the multiplication of two 64 bit numbers. This method allows us to design a multiplier of any bit number. From the comparative analysis it can be seen that the proposed multiplier has a better delay than previously designed multiplier of a same bit number. As our aim was to reduce the delay for 64 bit multiplier and in this case it is found to be 15.770 ns, so we have achieved our aim. Now one can proceed this work to make the 128 bit multiplier using a Vedic algorithm.

REFERENCES

- [1] V. Buddhé, P.M.Palsodkar, P.P.Palsodkar "Design and Verification of Dadda Based Binary floating Point multiplier", *IEEE International Conference on Communication and Signal Processing*, (ICCSP-2014), 2014.
- [2] P.Singh, B.Kakani, "Performance Comparison of Floating point multipliers by using different multiplication algorithm", *International Journal of Electronics and Communication (IJEC)* Volume 3, Issue 1, January 2015.
- [3] Jagadguru Swami Sri Bharati Krsna Tirthji Maharaja," Vedic Mathematics", Motilal Banarsidas, Varanasi, India, 1986.

- [4] R. K. Kodali, S. S. Yenamachintala, and L. Boppana, "FPGA implementation of vedic Floating Point Multiplier," *Signal Processing Informatics Communication and Energy Systems (SPICES), 2015 IEEE International Conference on*, 19-21 Feb 2015, pp. 1-4.
- [5] Ramalatha, M.Dayalan, K D Dharani, P Priya, and S Deoborah, "High speed energy efficient ALU Design using Vedic multiplication techniques", *IEEE Int. Conf. on Advances in Computational Tools for Engineering Applications (ACTEA-2009)*, pp. 600-603, July 15-17, 2009.
- [6] Garima Rawat, Khyati Rathore, Siddharth Goyal, Shefali Kala and Poornima Mittal, "Design and Analysis of ALU: Vedic Mathematics Approach", *International Conference on Computing, Communication and Automation (ICCCA2015)*, ISBN:978-1-4799-8890-7/15/©2015 IEEE.
- [7] Pushpalata Verma, K. K., Mehta, "Implementation of an Efficient Multiplier based on Vedic Mathematics Using EDA Tool", *International Journal of Engineering and Advanced Technology (IJEAT)* Vol. 1, Issue 5, ISSN: 2249-8958, June 2012
- [8] Honey Durga Tiwari, Ganzorig Gankhuyag, Chan Mo Kim and Yong Beom Cho, "Multiplier design based on ancient Indian Vedic Mathematician", *IEEE International SoC Design Conference Busan, South Korea*, pp. 65- 68, 2008.
- [9] S.S.Mahakalkar, S.L.Haridas, "Design of High Performance IEEE754 Floating point multiplier using Vedic mathematics", *2014 Sixth International Conference on Computational Intelligence and Communication Networks*, 978-1-4799-6929-6.
- [10] Prabir Saha, Arindam Banerjee, Partha Bhattacharyya, Anup Dandapat, "High Speed ASIC Design of Complex Multiplier Using Vedic Mathematics", in *proceedings IEEE students on technology symposium* 14-16 January, 2011, IIT Kharagpur.
- [11] Deodhe, Yeshwant, Sandeep Kakde, and Rashmi Deshmukh. "Design and Implementation of 8-Bit Vedic Multiplier Using CMOS Logic." In *Machine Intelligence and Research Advancement (ICMIRA), 2013 International Conference on*, pp. 340-344. *IEEE*, 2013.
- [12] Kakde ,Sandeep, Mithilesh Mahindra, Atish Khobragade, and Nikit Shah. "FPGA Implementation of 128-Bit Fused Multiply Add Unit for Crypto Processors." In *Security in Computing and Communications*, pp. 78-85. Springer International Publishing, 2015.
- [13] Bhoyar Rupali, Prasanna Palsodkar, and Sandeep Kakde. "Design and implementation of goldschmidts algorithm for floating point division and square root." In *Communications and Signal Processing (ICCSP), 2015 International Conference on*, pp. 1588-1592. *IEEE*, 2015
- [14] Mahendra , Mithilesh, Sandeep Kakde, and G. Somulu. "HDL implementation of 128-bit Fused Multiply Add unit for multi mode SoC." In *Communications and Signal Processing (ICCSP), 2014 International Conference on*, pp. 653-656. *IEEE*, 2014.